

Engineering Mechanics (SOF)

Batch: DS (1A)

Last Date of Submission: 23-09-2026

Full Marks: 30

1. Show that the vector $i + j + k$ is equally inclined to the coordinate axes. Find the common value of the angle of inclination in degrees. (1 mark)
2. Write down the direction ratios of the vector $i - j + 2k$ and hence calculate its direction cosines. (1 mark)
3. If the position vectors of the vertices of a triangle are $2i + 4j - k$, $4i + 5j + k$ and $3i + 6j - 3k$, what can you say about the nature of the triangle? (2 marks)
4. The two adjacent sides of a parallelogram are $2i + 4j - 5k$ and $i + 2j + 3k$. Find the unit vectors along its two diagonals. (2 marks)
5. Find a unit vector of magnitude 2.5 units which is parallel to the vector $3i + 4j$. (1 mark)
6. If the vectors $2i - j + k$, $i + 2j - 3k$ and $3i + aj + 5k$ are coplanar, find the value of a . (2 marks)
7. A unit vector $\mathbf{a}$ makes an angle of 45 degrees with the Z-axis. If $\mathbf{a} + i + j$ is a unit vector, find $\mathbf{a}$. (4 marks)
8. (See below, 2 marks)

The number of integral values of p for which

$(p+1)\hat{i} - 3\hat{j} + p\hat{k}$, $p\hat{i} + (p+1)\hat{j} - 3\hat{k}$ and

$-3\hat{i} + p\hat{j} + (p+1)\hat{k}$ are linearly dependent vectors is

(a) 0

(b) 1

(c) 2

(d) 3

9. Find the projection of $2i + 3j - 2k$ on $i + 2j + 3k$ and express it as a vector. (2 marks)
10. (See below, 2 marks)
If $|\mathbf{a}| + |\mathbf{b}| = |\mathbf{c}|$ and $\mathbf{a} + \mathbf{b} = \mathbf{c}$, then find the angle between $\mathbf{a}$ and $\mathbf{b}$.
11. (See below, 2 marks)

A particle is acted upon by constant forces $4\hat{i} + \hat{j} - 3\hat{k}$ and $3\hat{i} + \hat{j} - \hat{k}$ which displace it from a point $\hat{i} + 2\hat{j} + 3\hat{k}$ to the point $5\hat{i} + 4\hat{j} + \hat{k}$. Find the work done by the forces in standard units.

12. Find a unit vector perpendicular to the vectors $6i + 2j + 3k$ and $3i - 6j - 2k$. (2 marks)
13. Find the area of the triangle whose 2 adjacent sides are given by the vectors $-2i - 5k$ and $i - 2j - k$. (2 marks)
14. Suppose $\mathbf{a}$ is a given vector. Prove that $i \times (a \times i) + j \times (a \times j) + k \times (a \times k) = 2\mathbf{a}$. (2 marks)
15. (See below, 3 marks)

Find the vector of length 3 unit which is perpendicular to $\hat{i} + \hat{j} + \hat{k}$ and lies in the plane of $\hat{i} + \hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j}$.